KAPASA MAKASA UNIVERISTY

Course Outline: Introduction to Information and Communication Technology (ICT 100)

Course Information

Course Title: Introduction to Information and Communication Technology (ICT)
Course Code: ICT 100
Level: Undergraduate (Year 1)
Credits: 300
Prerequisite: None

Course Description

This course introduces students to the fundamental concepts of Information and Communication Technology. It explores the role of ICT in society, basic computer hardware and software, computer networks, the Internet, cybersecurity, and emerging technologies. The course aims to build digital literacy and equip students with essential ICT skills applicable in academics, careers, and daily life.

Course Objectives

- Explain the basic concepts and components of ICT.
- Identify types of computer hardware, software, and their functions.
- Understand the basics of computer networks and Internet services.
- Use word processing, spreadsheet, and presentation software effectively.
- Recognize the impact of ICT on society and ethical issues.
- Demonstrate awareness of cybersecurity threats and safe online practices.
- Explore emerging technologies and their implications.

Weekly Breakdown

Week	Topic	Content
1	Introduction to ICT	Definition, Importance, History, ICT vs IT, Applications in various sectors
2	Types of Computers and Their Components	Classification, Hardware overview (input, output, storage, CPU), peripherals
3	Software	System vs Application

		software, Operating systems, Utility programs
4	Number Systems & Data Representation	Binary, decimal, octal, hexadecimal systems, data types, encoding
5	Introduction to Operating Systems	Types of OS (Windows, Linux, MacOS), File management, User interface
6	Word Processing	Features of MS Word or LibreOffice Writer, editing, formatting, tables, references
7	Spreadsheets	Introduction to MS Excel, formulas, functions, charts, data management
8	Presentation Software	MS PowerPoint/LibreOffice Impress, slide creation, multimedia integration
9	Internet and Web Technologies	Browsers, Search engines, Web 1.0, 2.0, 3.0, email, cloud computing
10	Computer Networks	LAN, WAN, network topologies, protocols (TCP/IP), devices (router, switch)
11	Cybersecurity	Threats (viruses, phishing, hacking), safe practices, antivirus, backups
12	ICT in Society	Education, health, governance, commerce, digital divide, e-waste
13	Emerging Technologies	AI, IoT, cloud computing, blockchain, big data, 5G
14	Revision & Practical Assessment	Review of key concepts, hands-on tasks in lab

Assessment Methods

- Practical assessment: 10%
- Assignments (Theory and Practical): 20%
- Project/Group Presentation: 10%
- Final Exam (Theory and Practical): 60%

Recommended Textbooks (2015–present)

1. Parsons, J. J., & Oja, D. (2019). *New Perspectives on Computer Concepts 2019: Comprehensive*. Cengage Learning.
2. Shelly, G. B., Vermaat, M. E. (2017). *Discovering Computers 2018: Digital Technology, Data, and Devices*. Cengage Learning.
3. Goel, A. (2016). *Computer Fundamentals*. Pearson Education India.
4. Ray, C. (2020). *Introduction to Information Technology*. Jones & Bartlett Learning.
5. Laudon, K. C., & Laudon, J. P. (2021). *Management Information Systems: Managing the Digital Firm* (17th ed.). Pearson.
6. UNESCO/Intel/Microsoft Digital Literacy Curriculum: https://digital-literacy.org